
Autonomous Production Choke Controller

Reference Dataset Comparison

Honeywell Hackathon 2026

Engineering Submission

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Important Compliance Statement

The Honeywell reference dataset was used ONLY as a post-hoc engineering consistency check. It was NOT used for: system identification, dynamic model estimation, controller tuning, parameter optimization, controller design, or validation metric generation.

This complies with the official problem statement: 'Students are expected to generate their own data using the simulator and develop their control-oriented models from these experiments.'

Data Sources

Table 1: Data sources and their roles.

Source	Purpose	Usage
TestSimulator (built-in)	System identification	6 step tests, 90 calls
TestSimulator (built-in)	Controller validation	3 scenarios, 153 calls
TestSimulator (built-in)	Controller tuning	Config defaults, no fitting
Honeywell reference CSV	Post-hoc comparison	scratch/analyze_dataset.py
Honeywell reference CSV	Post-hoc comparison	scratch/full_analysis.py

Engineering Workflow

TestSimulator -> StepTestRunner (6 tests) -> ModelIdentifier (63.2% method) -> WellModel -> AutonomousChokeController -> 3 scenarios -> Validation results.

The reference CSV connects ONLY to scratch/ (standalone analysis, never imported by any pipeline module). There is no data path from the reference dataset to model identification, controller tuning, or validation.

Comparison Findings

Table 2: Qualitative consistency checks.

Check	Result
Gain sign convention	Consistent: Q positive, pressures negative
Operating range	Consistent: choke 0-100%, Q 0-200, pressures in expected ranges
Dynamic response shape	Consistent: FOPDT-like first-order responses
Noise characteristics	Consistent: Gaussian noise with plausible magnitudes

Table 3: Quantitative comparison (illustrative).

Parameter	TestSimulator	Reference (Observed)
Q gain [bbl/hr/%]	+2.0	Varies by regime
WHP gain [psi/%]	-3.0	Varies by regime
FLP gain [psi/%]	-2.0	Varies by regime
BHP gain [psi/%]	-4.0	Varies by regime

Note: The reference dataset was NOT used to set or tune any TestSimulator parameter. TestSimulator uses round-number engineering defaults chosen independently.

Conclusion

The simulator-generated data provides a complete and sufficient basis for controller development. The reference dataset serves its intended purpose: demonstrating that the engineering approach produces reasonable results consistent with expected well behavior. No reference dataset values were used in any controller design decision.